

A brief guide to statistical analysis of grouped data in preclinical research

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Clustering and nesting (C&N) arise in many preclinical studies, such as when animals are group-housed or share litters, or in cell culture. Ignoring C&N undermines the validity of analyses. Here, we explain how C&N arise, as well as valid designs and analyses.

Many published preclinical research analyses are not valid because they fail to consider the non-independence of data points sharing common conditions or influences (technically speaking, the non-independence of model residuals). For a thorough elucidation of the concept of non-independence, see ref. 1. Preclinical research often involves interventions delivered to groups, such as feeding a specific diet to a cage of mice or applying a treatment to a plate of cells. Even if interventions are not delivered to a group, clustering may still need to be accounted for².

We discuss why recognizing C&N is essential for valid analyses and substantiated results, discuss how C&N arise in common experimental models, and provide design and analysis guidance. Attention to these issues will improve the rigor and reproducibility of preclinical findings.

Defining clustering and nesting

Clustering refers to statistical dependence among observations within a group. This dependence is often due to shared environmental or genetic influences and can be positive (observations tend to vary together) or negative (observations tend to vary inversely)³. Nesting is a related concept describing a hierarchical structure: lower-level units (for example, individual animals or cells) are nested within higher-level units (for example, cages, litters or plates)³. Nesting needs to be accounted for if the unit of allocation is a group (for example, a cage of mice or a plate of cells), and clustering needs to be accounted for even without group allocation (Box 1).

Consequences of failing to account for C&N

In statistical terms, clustered or nested data violate the assumption of independence required by standard tests such as the *t*-test or ANOVA and so require other approaches³. Failing to do so can inflate the type I error rate, leading to artificially small standard errors, *P* values and associated confidence intervals. This can result in spurious statistical significance (that is, false positives), undermining preclinical findings^{4,5}.

Examples of C&N in preclinical research

C&N are pervasive in preclinical research (Fig. 1) and we now discuss some common scenarios.

Group-housed animals (cage effects). When animals are group-housed, such as in a cage, pen or chamber, they form a cluster. All animals

BOX 1

Glossary

Unit of allocation: The entity (for example, cage, litter or culture plate) assigned to treatment.

Experimental unit: The smallest distinct entity to which a treatment can be applied (for example, an individual animal or a cage).

Pseudoreplication: An error resulting from treating multiple measurements from a single experimental unit (for example, cells from the same animal) as independent replicates.

Random effect: A component in a statistical model that captures variation due to clustering.

Blocking factor: A known source of variability that cannot be eliminated but is accounted for during experimental design (such as during randomization) and statistical analysis to control its influence on the results.

Intraclass correlation coefficient (ICC): A statistic describing the similarity among observations within a group.

in one cluster experience the same environment (for example, bedding, temperature, light cycle, social interactions, microbiome exposure and infectious diseases⁶) and, consequently, their outcomes (body weight, glucose levels and so on) may be dependent (Fig. 1a). If animals already housed together within a cage are allocated to the same intervention (for example, a specific diet or a drug in their water), then the cage – not the individual animal – is the experimental unit, as animals are nested within cages. In this scenario, if there are five mice in each of two cages, there are not five independent mice per group; there are two independent experimental units (cages) per group, each containing five mice. At the extreme, if any treatment has only one group, it is not possible to separate the effect of the treatment and that of the group. In this case, valid statistical analyses cannot be performed⁷. This sometimes occurs if entire communities of animals are allocated to treatments, as is sometimes the case in research with naturalistic enclosures (for example, ref. 8). Clustering is a consideration regardless of whether the treatment is applied to the cage (such as a high-fat diet provided to the whole cage) or to the individual animals (such as an injection in group-housed mice, akin to an individually randomized group treatment design²). Whether the cage should be considered the experimental

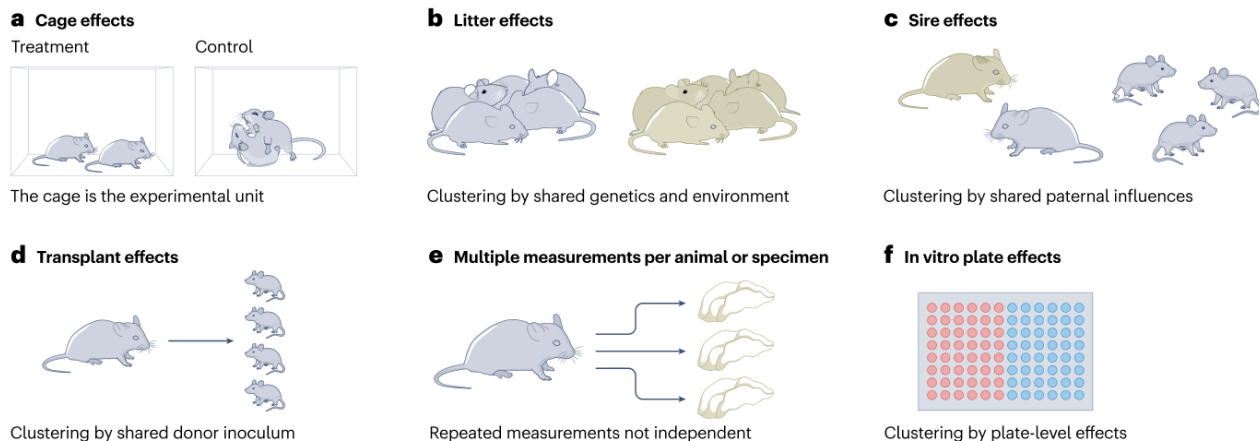


Fig. 1 | Common clustering and nesting scenarios in preclinical research.

a. Mice housed in the same cage experience shared conditions (for example, bedding, light cycle and microbiome), making the cage the cluster or experimental unit if a group-level treatment (such as diet) is applied. **b.** Offspring from the same dam share a genetic background and maternal environment, creating correlated outcomes within the litter. The litter can serve as a cluster or experimental unit when the intervention is applied to the pregnant dam. **c.** In studies in which multiple sires are used and each is mated to multiple dams, offspring may be clustered by sire owing to shared genetic contributions. This is common in genetic and livestock studies. Clustering arises only when offspring

from multiple sires are analysed together and some sires contribute to more than one litter. **d.** When biological material (for example, caecal contents or microbiota) from a limited number of donors is transplanted into multiple recipients, the recipients are clustered by donor. **e.** Repeated measures (for example, multiple tissues or time points) are nested within the same individual, meaning these data points are not independent and require hierarchical or repeated-measures approaches. **f.** Multiple wells on the same plate often share reagent batches and incubation conditions, inducing plate-level clustering. Each plate may act as a blocking factor or random effect in the analysis.

unit here depends on how time and grouping factors are reflected in the analysis, and more work is needed to determine guidance for such scenarios.

Invertebrates and small animals. Studies in invertebrates (such as *Drosophila* or *Caenorhabditis elegans*) often use group-based assays; for example, a feeding or preference test in which dozens of flies are tested together in a single vial or chamber. The readout might be a single metric for that group (for example, the proportion of flies responding to a stimulus). Here, as for mice in cages, the group or vial is the experimental unit – not each individual fly. In one analysis of a fly T-maze assay, using a *t*-test on n = approximately 100 flies per group (when in reality there was 1 group of approximately 100 flies per treatment) produced “extremely optimistic *P* values” for tiny differences in behaviour⁹. Similarly, if fish are allocated to tanks, the tank is often the experimental unit, not the individual fish. Hurlbert showed that nearly half of a sample of ecological field experiments that applied inferential statistics made the mistake of failing to account for C&N, often called pseudoreplication¹⁰.

Litter, sire and donor effects. In studies using offspring, using multiple siblings from the same litter in a study can introduce litter variability as a clustering factor (Fig. 1b). All pups in a litter share a mother and early-life environment. If a treatment is administered to a pregnant dam (for example, a maternal diet or drug), then the litter is the experimental unit, as pups are nested in the litter. Lazic et al. found that nearly half of a random sample of animal experiments with interventions applied to parents and outcomes measured in offspring did not appropriately consider litter effects¹¹. In addition, in multigeneration studies, particularly with livestock, when S sires are mated to D dams with $1 < S < D$ and offspring are studied, the offspring may be clustered

by sire from genetic or other transmissible factors¹² (Fig. 1c). Similarly, transplant studies (including fecal transplant) in which M (subject) mice receive transplants from m (donor) mice, with $1 < m < M$, can induce clustering (Fig. 1d).

Multiple measurements per animal. When multiple data points come from the same animal, those observations are nested within that animal (Fig. 1e). Examples include longitudinal repeated measures (such as body weight measured weekly for each animal), multiple tissue samples or organs analysed from one animal, or measurements on multiple cells or sections from one specimen. All these within-animal observations are not independent replicates; the animal is the experimental unit. For example, if each mouse contributes two measurements from the same tumour, those two measurements are nested within the mouse.

In vitro experiments

In vitro experiments (for example, measuring oxygen consumption or glucose uptake in cultured cells) are prone to clustering¹¹ (Fig. 1f). A common design is to have multiple wells per treatment on multi-well plates from the same biological sample (that is, nested, technical replicates). Shared plate conditions (temperature variations, edge effects or reagent batches) can cause dependence. If all treatment wells are on one plate and all control wells on another, plate-to-plate variability can confound the results. Even when treatments are mixed on a plate, measurements from the same plate may be dependent.

Site, batch and researcher effects. Data can also be nested by laboratory site (in multi-site studies), by experimental batch (for example, different days or assay runs), or by the researcher performing experiments or collecting data¹³.

Comment

Best practices when planning and analysing for C&N

Careful planning may involve recognizing and reducing C&N at the design stage or choosing appropriate statistical models to account for these effects. Housing animals individually can eliminate cage-level clustering but is often not possible for ethical, social or treatment-related reasons. When group-housing is required, the group (if allocated that way) must be treated as the experimental unit in both power calculations and analyses. Even if treatments are administered individually within a group, it is always important to account for sources of clustering, for instance by including the group as a random effect or by aggregating outcomes at the group level for valid analyses¹⁴. If animals are housed alone but were originally group-housed, we recommend that group be included in statistical models given that clustering is plausible. Furthermore, stratified randomization at the design stage, using the origin group as a blocking factor, makes group membership and treatment condition orthogonal, which may reduce the variance of the estimated treatment effect and increase power. Additional considerations exist for statistical power when the group is the experimental unit. Although increasing the number of animals per group and the number of groups can each increase statistical power, as a general rule of thumb, increasing the number of groups will increase statistical power much more so than increasing the number of animals per group for any given number of animals needed¹⁵. This is especially true when the number of animals is large and the number of groups is relatively small.

If litter effects are possible, even if treatments are given post-weaning to individual offspring, modelling the litter as a random effect can address non-independence¹⁴. Sire and dam should also be included as random effects¹². When feasible, use only one animal per litter per group. If that is not practical, balancing litter representation across groups using stratified randomization helps control litter-to-litter variability.

Similar considerations apply to cell culture experiments and plate effects. The plate (or experimental run or day) is a blocking factor or random effect, and multiple measurements from the same donor do not represent independent observations. Thus, the sample size for statistical tests is the number of independent experiments (for example, animals or culture preparations), rather than the number of wells or cells. Plate effects can be addressed by randomizing treatment allocation across plates, including plate as a random factor in analysis, or by averaging results per plate or per donor before comparison.

Other nesting factors (for example, site, batch or researcher effects) can be addressed during experimental design. Blocking designs offer a strategy to account for these (“control what you can, block what you cannot, and randomize the rest”)¹³.

When analysing clustered or nested designs, two approaches are (1) aggregate, then analyse, or (2) analyse with mixed or hierarchical models. The first approach means reducing the data from each cluster to a single summary (for example, mean per cage) and then using those summaries as independent observations in a traditional model (for example, *t*-test or ANOVA). However, this approach loses information, such as the within-cluster variability, which can be of interest. The second approach uses all data but includes cluster as a random effect or as a blocking factor in a multi-level model. For example, a linear mixed-effects model can include a random intercept for each cage or litter. This correctly partitions variance into within-cluster and between-cluster components and, when appropriately specified, adjusts degrees of freedom to account for nesting of clusters within

treatment, as degrees of freedom for analyses must be based on the independent experimental unit³. Another option is using generalized estimating equations or cluster-robust standard errors if sample sizes of experimental units are large enough, although required sample sizes are typically much larger than is common in preclinical research³. The key point is that the analysis must reflect the data hierarchy: measurements nested within animals, animals within cages and so on, and the observations dependent on the interactions of experimental units.

Investigators are encouraged to follow reporting guidelines specific to their research approach, such as ARRIVE for in vivo animal research¹⁴, and report statistical model details (for example, random effects, experimental unit, degrees of freedom adjustments) and grouping factor intraclass correlation coefficients (ICCs). Good communication between non-statistician and statistician researchers is essential so that each understands the research question, experimental design and ethical considerations from planning to analysis.

Summary and conclusions

We recommend (1) avoiding designing studies with grouping effects when possible, although often they are unavoidable; (2) identifying and reporting grouping effects and ICCs transparently, following relevant reporting guidelines; and (3) regardless of ICC, always including grouping factors in analyses using suitable methods and adjusting degrees of freedom when necessary. C&N are concepts common to preclinical designs, and accounting for them in design and analysis phases is essential. It is an ethical imperative that research be designed and analysed to avoid wasting investment of animals, research dollars and effort.

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Author contributions

C.J.V. and D.B.A. conceptualized the manuscript. C.J.V. prepared the original draft. C.J.V., L.G.A. and D.B.A. wrote, edited and revised the manuscript.